



1025512576 Sba301 Pe Huongdan

Software Architecture And Design (Trường Đại học FPT)



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HUONG DAN GIAI DE THI PE

SBA301 - Integrate Single Page Application with Spring Boot

Shoes Management System (SMS) | React JS + Spring Boot

PHAN 1: TONG QUAN DE THI

Thông tin	Chi tiết
Subject Code	SBA301
Subject Name	Integrate Single Page Application with Spring Boot
Thời gian	90 phút (bao gồm nộp bài)
Loại	PE Practical Exam (Trial)
Project Code	SMS

Lưu ý quan trọng tu đề

- Được kết nối Internet trong 10 phút đầu để setup project
- Chỉ được dùng tài nguyên trên máy local sau đó
- PHẢI chỉnh sửa backend để cho phép kết nối từ ReactJS app (CORS)

Tóm tắt chức năng cần làm

Trang	URL	Chức năng chính
Shoes List	/	Hiển danh sách, filter, phân trang, xóa
Add New	/add	Form thêm mới shoes với validation
View Details	/detail/:id	Xem chi tiết 1 đôi shoes

PHAN 2: CAU HINH BACKEND (Spring Boot)

Buoc 1: Them CORS Config

Tao file CorsConfig.java trong package config của backend:

```
package fu.sba301.pe2026.config;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.web.servlet.config.annotation.CorsRegistry;
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;

@Configuration
public class CorsConfig {
    @Bean
    public WebMvcConfigurer corsConfigurer() {
        return new WebMvcConfigurer() {
            @Override
            public void addCorsMappings(CorsRegistry registry) {
                registry.addMapping("/api/**")
                    .allowedOrigins("http://localhost:5173")
                    .allowedMethods("GET", "POST", "PUT", "DELETE")
                    .allowedHeaders("*");
            }
        };
    }
}
```

! Sau khi them file nay, restart Spring Boot de ap dung thay doi.

Buoc 2: Kiem tra application.properties

Dam bao cac config nay chinh xac khi nop bai:

```
spring.datasource.url=jdbc:sqlserver://localhost:1433;databaseName=SBA301_2026_PE
spring.datasource.username=sa
spring.datasource.password=sa
server.port=8080
```

Tom tat REST API của Backend

#	Path	Method	Mo ta
1	/api/shoes/search	GET	Tim kiem shoes (filter + phan trang)
2	/api/shoes	POST	Tao moi shoes
3	/api/shoes/{id}	GET	Lay chi tiet 1 shoes
4	/api/shoes/{id}	DELETE	Xoa shoes theo id
5	/api/categories	GET	Lay danh sach category

Cau truc Response tu API

ShoesResponse (du lieu 1 shoes tra ve):

```
{
  "shoesId": 1,
  "shoesName": "Nike Air Max",
  "price": 150.0,
  "quantity": 50,
  "manufacturer": "Nike",
  "productionDate": "2025-01-20T00:00:00.000+00:00",
  "importDate": "2025-01-25T00:00:00.000+00:00",
  "categoryName": "Sport"
}
```

PageResponse (response phan trang):

```
{
  "content": [...],
  "currentPage": 0,
  "pageSize": 5,
  "totalElements": 10,
  "totalPages": 2,
  "first": true,
  "last": false
}
```

ShoesRequest (data gui len khi tao moi):

```
{
  "shoesName": "Nike Air Max",
  "price": 150.0,
  "quantity": 1,
  "manufacturer": "Nike",
  "productionDate": "2025-01-20T00:00:00.000Z",
  "importDate": "2025-01-25T00:00:00.000Z",
  "categoryId": 1
}
```

PHAN 3: SETUP PROJECT FRONTEND

Buoc 1: Tao project voi Vite

```
npm create vite@latest

# Tra loi cac cau hoi:
# Project name: <ClassName>_<StudentID>_SMS_PE
# Framework: React
# Variant: JavaScript
```

! Ten folder PHAI theo convention: <ClassName>_<StudentID>_<ProjectCode>_PE

Tip: Vi du: SE1873_HE180000_SMS_PE

Buoc 2: Cai cac thu vien can thiet

```
cd <ten-folder-vua-tao>
npm install
npm install react-router-dom axios react-bootstrap bootstrap
```

Buoc 3: Tao file jsconfig.json (BAT BUOC theo de)

Tao file jsconfig.json tai root folder (cung cap voi package.json):

```
{
  "compilerOptions": {
    "baseUrl": ".",
    "paths": {
      "*": ["../../node_modules/*"]
    }
  }
}
```

Buoc 4: Cau truc thu muc src

```
src/
|-- components/
|   |-- Header.jsx      <- Logo + tieu de + ngay thang
|-- pages/
|   |-- ShoesList.jsx   <- Trang danh sach (trang chu)
|   |-- ShoesDetail.jsx <- Trang xem chi tiet
|   |-- AddShoes.jsx    <- Trang them moi
|-- services/
|   |-- api.js          <- Tat ca ham goi API
|-- App.jsx             <- Cau hinh routes
|-- main.jsx            <- Entry point
```

PHAN 4: CODE TUNG FILE

File: src/main.jsx

Entry point của app. Import Bootstrap CSS tại đây.

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import 'bootstrap/dist/css/bootstrap.min.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

File: src/App.jsx

Khai báo routes - URL nào thì hiển thị trang nào.

```
import { BrowserRouter, Routes, Route } from 'react-router-dom';
import ShoesList from './pages/ShoesList';
import ShoesDetail from './pages/ShoesDetail';
import AddShoes from './pages/AddShoes';

function App() {
  return (
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<ShoesList />} />
        <Route path="/add" element={<AddShoes />} />
        <Route path="/detail/:id" element={<ShoesDetail />} />
      </Routes>
    </BrowserRouter>
  );
}
export default App;
```

File: src/services/api.js

Tập trung tất cả hàm gọi API tại 1 chỗ. Chỉ cần sửa baseURL nếu backend đổi port.

```
import axios from 'axios';

const axiosInstance = axios.create({
  baseURL: 'http://localhost:8080',
  headers: { 'Content-Type': 'application/json' },
});

export const getAllCategories = () => {
  return axiosInstance.get('/api/categories');
};

export const searchShoes = (name='', category='', page=0, size=5) => {
  return axiosInstance.get('/api/shoes/search', {
    params: {
      name: name || undefined,
      category: category || undefined,
      page,
      size,
    },
  });
};
```

```

};

export const getShoesById = (id) => {
  return axiosInstance.get(`/api/shoes/${id}`);
};

export const createShoes = (shoesData) => {
  return axiosInstance.post('/api/shoes', shoesData);
};

export const deleteShoes = (id) => {
  return axiosInstance.delete(`/api/shoes/${id}`);
};

```

File: src/components/Header.jsx

Component tai su dung o tat ca cac trang. Nhan prop 'title' de hien tieu de.

```

import { useState, useEffect } from 'react';

function Header({ title }) {
  const [currentDate, setCurrentDate] = useState('');

  useEffect(() => {
    const now = new Date();
    const day = String(now.getDate()).padStart(2, '0');
    const month = String(now.getMonth() + 1).padStart(2, '0');
    const year = now.getFullYear();
    setCurrentDate(`${day}/${month}/${year}`);
  }, []);

  return (
    <div style={{ borderBottom: '2px solid #dee2e6', marginBottom: '20px' }}>
      <div className='d-flex justify-content-between align-items-center px-3 pt-2'>
        <div style={{ border: '1px solid #000', padding: '4px 12px', fontWeight:
'bold' }}>
          Logo
        </div>
        <div>Date: {currentDate || 'dd/MM/yyyy'}</div>
      </div>
      <div className='text-center mt-1'>
        <strong style={{ fontSize: '18px' }}>{title}</strong>
      </div>
    </div>
  );
}
export default Header;

```


File: src/pages/ShoesList.jsx

Trang phục tạp nhất: danh sách, filter, phân trang, xóa với confirm dialog.

Các state cần khai báo:

State	Kiểu	Mục đích
shoesList	Array	Danh sách shoes hiển thị trên bảng
categories	Array	Danh sách category cho dropdown filter
filterName	String	Giá trị nhập vào tìm theo tên
filterCategory	String	Category đang chọn trong dropdown
currentPage	Number	Trang hiện tại (bắt đầu từ 0)
pageInfo	Object	Thông tin phân trang từ API
showDeleteModal	Boolean	Hiện/ẩn dialog xác nhận xóa
shoesToDelete	Object	Shoes đang được chọn để xóa
message	String	Thông báo thành công/lỗi

Logic quan trọng:

- `useEffect([])` - gọi `fetchCategories()` 1 lần khi trang load
- `useEffect([currentPage])` - gọi `fetchShoes()` mỗi khi đổi trang
- `handleFilter()` - reset `currentPage` về 0, gọi lại `fetchShoes()`
- `handleDeleteClick(shoes)` - lưu shoes vào state, hiện modal
- `handleConfirmDelete()` - gọi API xóa, hiện thông báo, load lại list

Format date từ API (timestamp -> dd/MM/yyyy):

```
const formatDate = (dateValue) => {
  if (!dateValue) return '';
  const date = new Date(dateValue);
  const day = String(date.getDate()).padStart(2, '0');
  const month = String(date.getMonth() + 1).padStart(2, '0');
  const year = date.getFullYear();
  return `${day}/${month}/${year}`;
};
```

Render STT dung khi có phân trang:

```
// STT = số trang hiện tại x số item/trang + index + 1
{pageInfo.currentPage * pageInfo.pageSize + index + 1}
```

Render Pagination:

```
const renderPagination = () => {
  if (pageInfo.totalPages <= 1) return null;
  const items = [];
  items.push(
    <Pagination.Prev key='prev'
      disabled={pageInfo.currentPage === 0}
      onClick={() => handlePageChange(currentPage - 1)} />
  );
  for (let i = 0; i < pageInfo.totalPages; i++) {
```

```
items.push(  
  <Pagination.Item key={i}  
    active={i === pageInfo.currentPage}  
    onClick={() => handlePageChange(i)}>  
    {i + 1}  
  </Pagination.Item>  
);  
}  
items.push(  
  <Pagination.Next key='next'  
    disabled={pageInfo.last}  
    onClick={() => handlePageChange(currentPage + 1)} />  
);  
return <Pagination className='justify-content-end'>{items}</Pagination>;  
};
```

File: src/pages/ShoesDetail.jsx

Hien thi chi tiet 1 shoes. Lay id tu URL bang useParams().

```
import { useState, useEffect } from 'react';
import { useParams, useNavigate } from 'react-router-dom';
import { Container, Button, Card, Row, Col } from 'react-bootstrap';
import Header from '../components/Header';
import { getShoesById } from '../services/api';

function ShoesDetail() {
  const { id } = useParams(); // Lay :id tu URL /detail/:id
  const navigate = useNavigate();
  const [shoes, setShoes] = useState(null);

  useEffect(() => {
    getShoesById(id).then(res => setShoes(res.data));
  }, []);

  if (!shoes) return <div>Loading...</div>;

  return (
    <Container>
      <Header title='View Details' />
      <Card className='mx-auto mt-3' style={{ maxWidth: '500px' }}>
        <Card.Body>
          <Card.Title className='text-center'>VIEW DETAILS</Card.Title>
          <Row className='mb-2'>
            <Col md={5} className='text-end fw-bold'>Shoes Name:</Col>
            <Col md={7}>{shoes.shoesName}</Col>
          </Row>
          // ... cac Row khac tuong tu ...
          <div className='text-center mt-4'>
            <Button variant='secondary' onClick={() => navigate('/')}>
              Quay Lai
            </Button>
          </div>
        </Card.Body>
      </Card>
      <p className='text-center mt-4'>©Copyright 2026</p>
    </Container>
  );
}

export default ShoesDetail;
```

File: src/pages/AddShoes.jsx

Form them moi shoes voi day du validation theo de.

Validation rules theo de:

Field	Bat buoc	Rule
Shoes Name	Co	Toi da 100 ky tu, khong trung
Price	Co	La so, > 0 va < 10000
Manufacturer	Co	Toi da 100 ky tu
Production Date	Co	Format dd/MM/yyyy
Import Date	Khong	Format dd/MM/yyyy
Category	Co	Chon tu dropdown, lay tu API

Luu y khi gui date len API:

```
// Input type='date' tra ve chuoi 'yyyy-MM-dd'
// Can convert sang Date object truoc khi gui
productionDate: formData.productionDate ? new Date(formData.productionDate) : null,

// Backend can field 'quantity' du de khong yeu cau nhap
quantity: 1, // Dat mac dinh = 1
```

Luong xu ly khi click Save:

- Buoc 1: Goi ham validate() kiem tra tat ca field
- Buoc 2: Neu co loi -> luu vao state errors, hien loi do, dung lai
- Buoc 3: Neu khong loi -> goi API createShoes(payload)
- Buoc 4: Hien thong bao 'Created new shoes successfully'
- Buoc 5: Reset form ve trong
- Buoc 6: Sau 2 giay -> navigate('/') ve trang danh sach

PHAN 5: CHECKLIST TRUOC KHI NOP BAI

Kiem tra chuc nang

	Chuc nang	Trang
[]	Hien du danh sach shoes khi vao trang	Shoes List
[]	Filter theo ten -> dung ket qua	Shoes List
[]	Filter theo category -> dung ket qua	Shoes List
[]	Khong co ket qua -> hien 'No records found'	Shoes List
[]	Pagination chuyen trang duoc	Shoes List
[]	Click Delete -> hien dialog xac nhan	Shoes List
[]	Click Yes -> hien 'Deleted successfully', list cap nhat	Shoes List
[]	Click Close -> dong dialog, khong xoa	Shoes List
[]	Click Add New -> chuyen sang form	Shoes List
[]	Save khi de trong -> hien loi do tung field	Add New
[]	Dien du -> Save -> 'Created new shoes successfully'	Add New
[]	Sau Save -> tu ve trang danh sach	Add New
[]	Click Back -> ve trang danh sach	Add New
[]	Click View -> hien dung thong tin chi tiet	View Details
[]	Ngay hien thi dung dd/MM/yyyy	View Details
[]	Click Quay Lai -> ve trang danh sach	View Details

Kiem tra cau hinh

	Yeu cau	Ghi chu
[]	Ten folder dung convention	<ClassName>_<StudentID>_SMS_P E
[]	Co file jsconfig.json o root folder	Bat buoc theo de
[]	App chay tai http://localhost:5173/	Port phai la 5173
[]	Ket noi backend port 8080	baseUrl trong api.js
[]	Backend co CorsConfig.java	Cho phep origin 5173
[]	application.properties dung config	DB: SBA301_2026_PE, sa/sa

Cac buoc nop bai

1. Xoa thu muc node_modules trong project React
2. Chuot phai vao folder project -> Send to -> Compressed (zipped) folder
3. Nop file .zip len EOS

! Backup project sang folder khac truoc khi xoa node_modules!

PHAN 6: LOI THUONG GAP VA CACH FIX

Loi	Nguyen nhan	Cach fix
does not provide an export named 'default'	File chua co export default	Them export default <TenFunction> o cuoi file
Comment <code>{/* */}</code> inline trong attribute JSX	Khong duoc comment cung dong voi attribute	Xoa comment inline, chi comment tren dong riêng
CORS error khi goi API	Backend chua co CorsConfig	Them CorsConfig.java vao backend, restart
Man hinh trang khi vao app	File page chua co noi dung	Them placeholder function + export default
Ngay hien thi sai format	API tra ve timestamp	Dung ham formatDate() de convert
Dropdown Category rong	Chua goi fetchCategories()	Them useEffect(() => fetchCategories(), [])

Loi quan trong: Comment inline trong JSX

! KHONG duoc viet comment cung dong voi attribute trong JSX!

SAI (gay loi parse):

```
<Form.Control
  name='shoesName'  {/* comment o day SE BI LOI */}
/>
```

DUNG (comment tren dong riêng):

```
{/* Comment phai nam ben TRONG the JSX */}
<Form.Control
  name='shoesName'
/>
```

Chuc ban thi tot!